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# Blockchain Lab EXP 3

**Aim -** Create a Cryptocurrency using Python and perform mining in the Blockchain created.

**Theory -**

1. **Blockchain Overview**

A blockchain is a **distributed, decentralized digital ledger** used to record transactions securely across multiple computers (nodes). Instead of relying on a central authority, every node in the network maintains its own copy of the blockchain.
Each block in the blockchain contains:

- A list of transactions
- A timestamp
- The hash of the previous block
- Its own cryptographic hash

The blocks are linked together using hashes, forming a chain. Any attempt to modify data in a block changes its hash, which breaks the chain and makes tampering easily detectable. This property ensures **data integrity, transparency, and immutability**.

**2. Mining**

Mining is the process by which new blocks are added to the blockchain. It involves:

- Collecting pending transactions
- Creating a block header
- Solving a computationally difficult problem known as **Proof-of-Work (PoW)**

In Proof-of-Work, miners repeatedly change a nonce value to generate a hash that satisfies the network's difficulty requirement (such as leading zeroes). Once a valid hash is found, the block is added to the blockchain and shared with other nodes. As an incentive, miners receive a **cryptocurrency reward** for their computational effort.

**3. Multi-Node Blockchain Network**

In this experiment, a **multi-node blockchain network** is simulated using three nodes running on different ports (5001, 5002, and 5003).
Each node:

- Operates independently
- Maintains its own copy of the blockchain
- Communicates with peer nodes to share newly mined blocks

This setup demonstrates how blockchain achieves decentralization and synchronization without a central server.

**4. Consensus Mechanism**
To ensure consistency across all nodes, the **Longest Chain Rule** is used as the consensus mechanism.
When different versions of the blockchain exist, the node accepts the **longest valid chain** as the correct one. This rule ensures that all nodes eventually agree on a single transaction history, even if temporary differences arise.

**5. Transactions and Mining Reward**
Transactions in the cryptocurrency system include:

- Sender address
- Receiver address
- Transaction amount

When a block is mined, all pending transactions are added to the block. Additionally, a special transaction is created to reward the miner with newly generated cryptocurrency. This reward mechanism motivates miners to maintain and secure the network.

**6. Chain Replacement**
The chain replacement process ensures network consistency. When the /replace_chain endpoint is triggered:

- A node requests blockchain data from its peers
- Validates the received chains
- Replaces its own chain if a longer and valid chain is found

This mechanism helps resolve conflicts and keeps all nodes synchronized.

**Code -**

```
class Blockchain:
    def __init__(self):
        self.chain = []
        self.transactions = []
        self.create_block(proof=1, previous_hash='0')
        self.nodes = set()
    def create_block(self, proof, previous_hash):
        block = {
            'index': len(self.chain) + 1,
```

```
            'timestamp': str(datetime.datetime.now()),
            'proof': proof,
            'previous_hash': previous_hash,
            'transactions': self.transactions
        }
        self.transactions = []
        self.chain.append(block)
        return block

    def get_previous_block(self):
        return self.chain[-1]

    def proof_of_work(self, previous_proof):
        new_proof = 1
        check_proof = False
        while not check_proof:
            hash_operation = hashlib.sha256(
                str(new_proof**2 - previous_proof**2).encode()
            ).hexdigest()
            if hash_operation[:4] == '0000':
                check_proof = True
            else:
                new_proof += 1
        return new_proof

    def hash(self, block):
        encoded_block = json.dumps(block, sort_keys=True).encode()
        return hashlib.sha256(encoded_block).hexdigest()

    def is_chain_valid(self, chain):
        previous_block = chain[0]
        block_index = 1
        while block_index < len(chain):
            block = chain[block_index]
            if block['previous_hash'] != self.hash(previous_block):
                return False
            previous_proof = previous_block['proof']
            proof = block['proof']
            hash_operation = hashlib.sha256(
                str(proof**2 - previous_proof**2).encode()
```

```
            ).hexdigest()
            if hash_operation[:4] != '0000':
                return False
            previous_block = block
            block_index += 1
        return True

    def add_transaction(self, sender, receiver, amount):
        self.transactions.append({
            'sender': sender,
            'receiver': receiver,
            'amount': amount
        })
        previous_block = self.get_previous_block()
        return previous_block['index'] + 1

    def add_node(self, address):
        parsed_url = urlparse(address)
        self.nodes.add(parsed_url.netloc)

    def replace_chain(self):
        network = self.nodes
        longest_chain = None
        max_length = len(self.chain)
        for node in network:
            response = requests.get(f'http://{node}/get_chain')
            if response.status_code == 200:
                length = response.json()['length']
                chain = response.json()['chain']
                if length > max_length and self.is_chain_valid(chain):
                    max_length = length
                    longest_chain = chain
        if longest_chain:
            self.chain = longest_chain
            return True
        return False

app = Flask(__name__)

node_address = str(uuid4()).replace('-', '')
```

```
blockchain = Blockchain()

@app.route('/mine_block', methods=['GET'])
def mine_block():
    previous_block = blockchain.get_previous_block()
    previous_proof = previous_block['proof']
    proof = blockchain.proof_of_work(previous_proof)
    previous_hash = blockchain.hash(previous_block)
    blockchain.add_transaction(
        sender=node_address,
        receiver='Richard',
        amount=1
    )
    block = blockchain.create_block(proof, previous_hash)
    response = {
        'message': 'Congratulations, you just mined a block!',
        'index': block['index'],
        'timestamp': block['timestamp'],
        'proof': block['proof'],
        'previous_hash': block['previous_hash'],
        'transactions': block['transactions']
    }
    return jsonify(response), 200

@app.route('/get_chain', methods=['GET'])
def get_chain():
    response = {
        'chain': blockchain.chain,
        'length': len(blockchain.chain)
    }
    return jsonify(response), 200

@app.route('/is_valid', methods=['GET'])
def is_valid():
    is_valid = blockchain.is_chain_valid(blockchain.chain)
    if is_valid:
        response = {'message': 'All good. The Blockchain is valid.'}
    else:
        response = {'message': 'The Blockchain is not valid.'}
```

```
    return jsonify(response), 200

@app.route('/add_transaction', methods=['POST'])
def add_transaction():
    json_data = request.get_json()
    transaction_keys = ['sender', 'receiver', 'amount']
    if not all(key in json_data for key in transaction_keys):
        return 'Some elements of the transaction are missing', 400
    index = blockchain.add_transaction(
        json_data['sender'],
        json_data['receiver'],
        json_data['amount']
    )
    response = {
        'message': f'This transaction will be added to Block {index}'
    }
    return jsonify(response), 201

@app.route('/connect_node', methods=['POST'])
def connect_node():
    json_data = request.get_json()
    nodes = json_data.get('nodes')
    if nodes is None:
        return "No node", 400
    for node in nodes:
        blockchain.add_node(node)
    response = {
        'message': 'All the nodes are now connected.',
        'total_nodes': list(blockchain.nodes)
    }
    return jsonify(response), 201

@app.route('/replace_chain', methods=['GET'])
def replace_chain():
    is_chain_replaced = blockchain.replace_chain()
    if is_chain_replaced:
        response = {
            'message': 'The chain was replaced by the longest one.',
            'new_chain': blockchain.chain
        }
```

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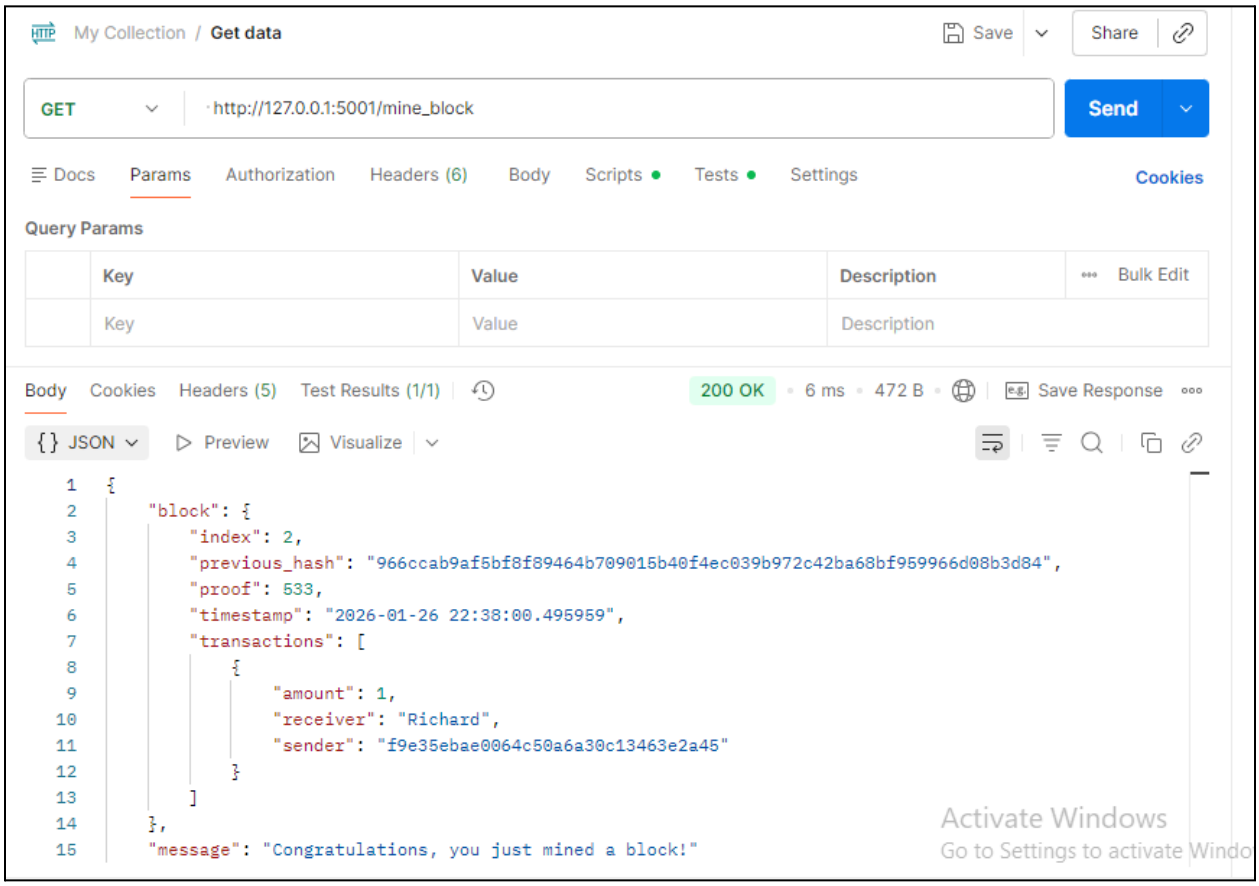
```
    else:
        response = {
            'message': 'The chain is already the largest one.',
            'actual_chain': blockchain.chain
        }
    return jsonify(response), 200

app.run(host='0.0.0.0', port=5000)
```



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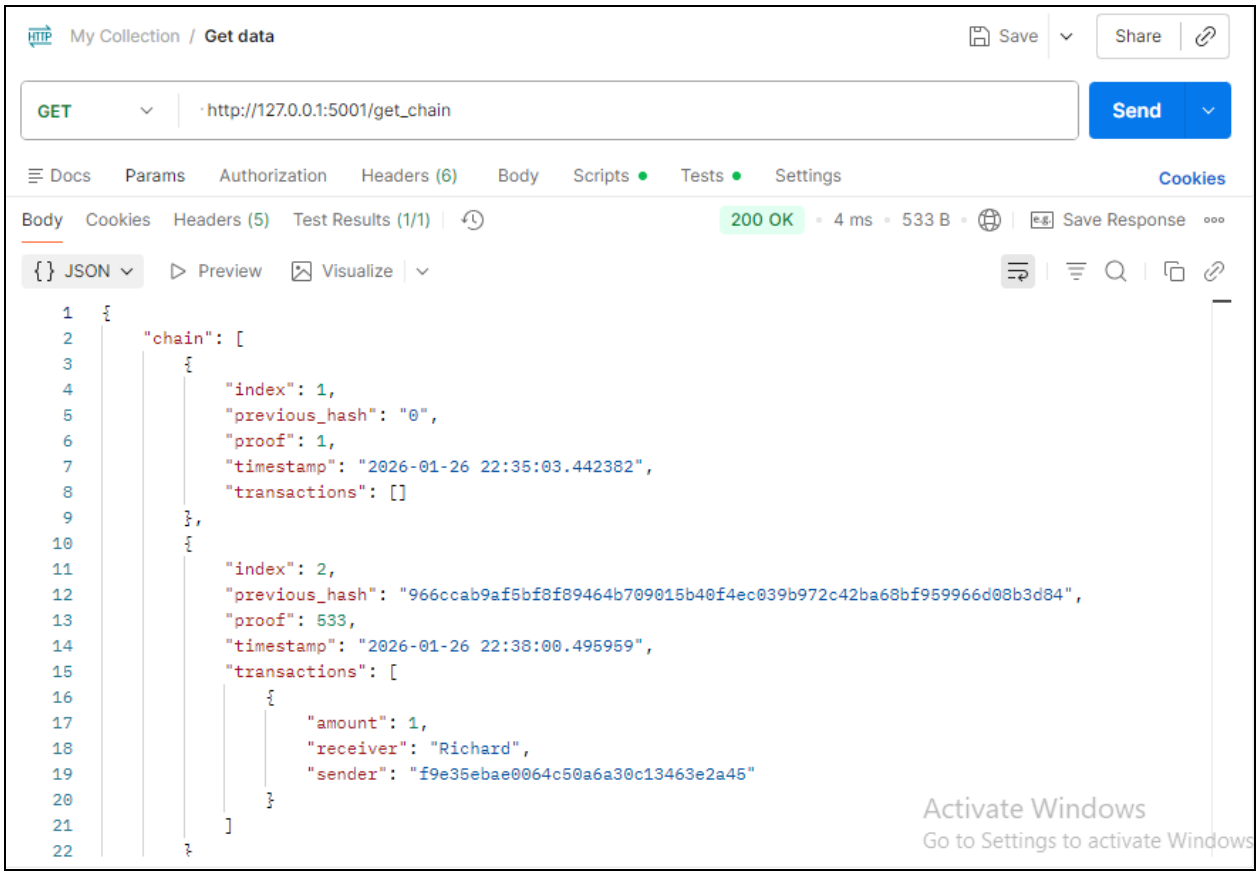


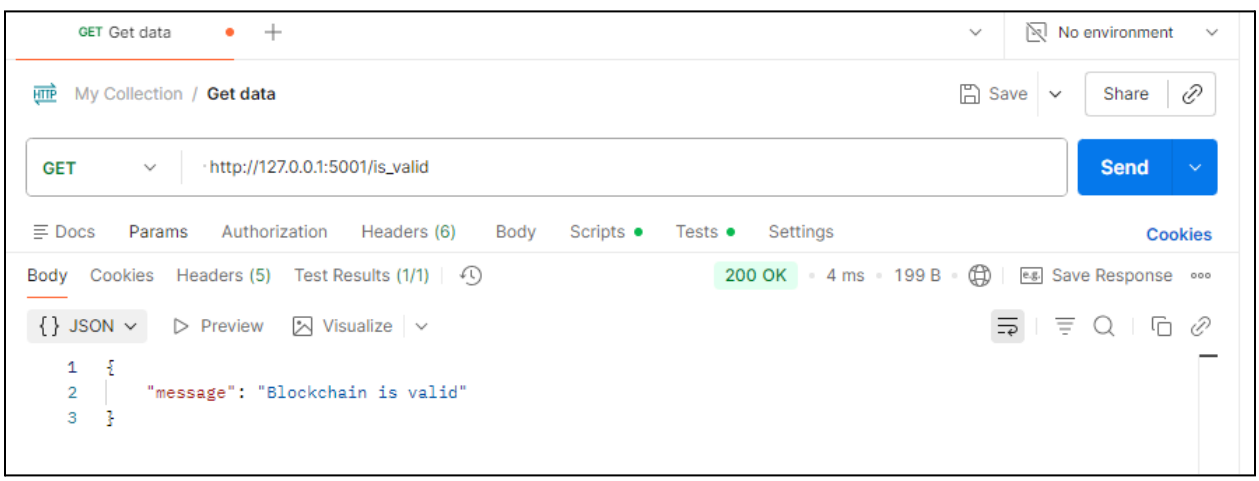


**Conclusion -** Learned to create cryptocurrency and mining process using python.